Artwork/Project Title
Satukerja

Year Accomplished
2023

Role/Position
Fullstack Developer

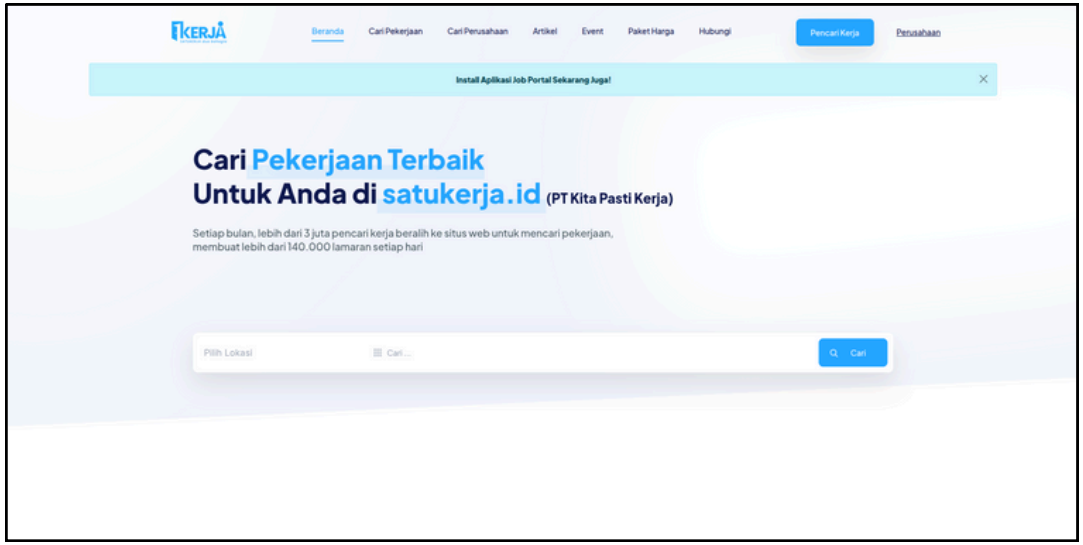
Publication Link
https://satukerja.id/

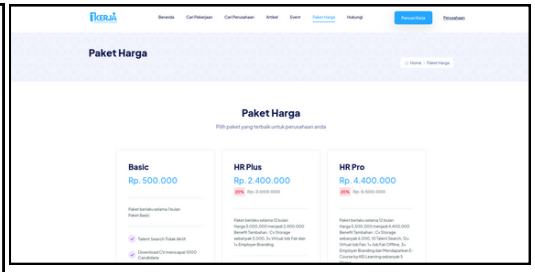
Artwork/Project Description

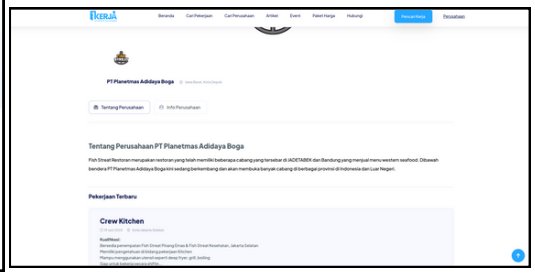
Group Project: Website For Streamlining the Hiring Process for Jobseekers and Employers

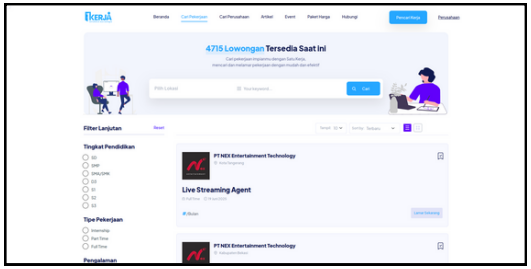
Satukerja was designed to streamline the entire hiring funnel: search, apply, communicate, and pay, by connecting employers and candidates on a single, intuitive portal. At the beginning of this project, I was a Laravel developer with no prior experience in the technologies we planned to use. Taking on the role of Full Stack Developer in a two-person team, I stepped far outside my comfort zone and committed myself to learning and mastering an entirely new stack from scratch. I dove deep into Next.js, understanding how server-side rendering works, how to build fast and SEO-friendly pages, and how to structure scalable frontend projects using Next.js. I learned Firebase step-by-step from how Firestore stores and retrieves data in real time, to implementing secure multi-provider authentication with Firebase Auth, and deploying Cloud Functions to handle backend logic without traditional servers. I selected and integrated libraries that balanced performance and flexibility, and carefully designed a Firestore schema with composite indexes to support dynamic sorting, filtering, and pagination. One of the most challenging parts was building complex queries within Firestore's constraints, such as filtering by salary range, location, and keywords simultaneously. I overcame this through extensive trial and error, denormalizing data structures, creating composite indexes, and implementing batched queries with client-side caching to preserve performance.

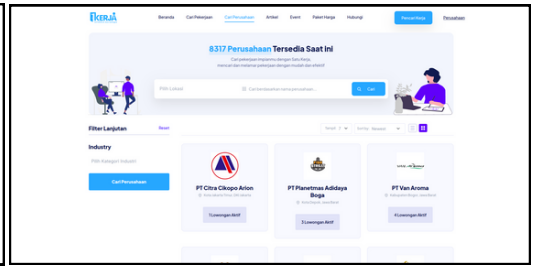
I built custom API routes in Next.js for features like email workflows using Nodemailer, and integrated a secure payment system using Midtrans, complete with webhook handling and reconciliation logic. On the frontend, I developed secure login flows, dynamic job feeds, an admin panel, and a CMS-style blog. Despite many moments of doubt and technical roadblocks, I kept pushing forward by breaking down problems, seeking feedback, and pair-programming with my teammate. As the project progressed, I not only became fluent in Next.js and Firebase, but also improved my ability to collaborate, adapt quickly, and prioritize user experience. In the end, I was able to significantly improve the website's performance, resulting in faster page loads and a smoother, more responsive interface. This project was more than just a technical milestone, it was a personal transformation. It taught me how to navigate uncertainty, learn independently, and thrive outside my comfort zone. Satukerja became a turning point in my journey as a developer, giving me the confidence and experience to tackle complex, user-centric web applications using modern tools and best practices.

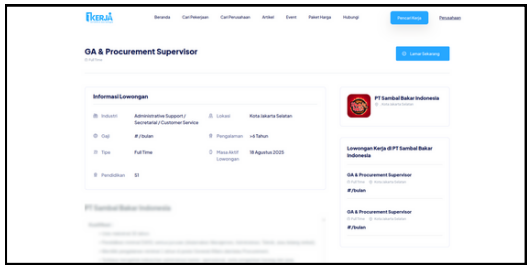












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Portfolio Submission for
Apple Developer Academy Cohort 2026

Artwork/Project Title
GMP CRM

Year Accomplished
2024

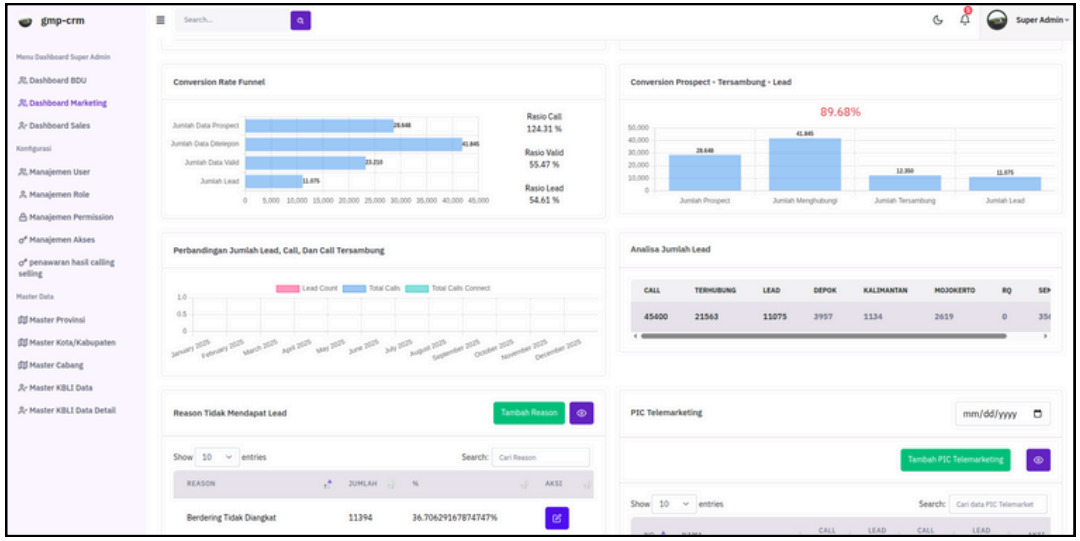
Role/Position
Project Manager

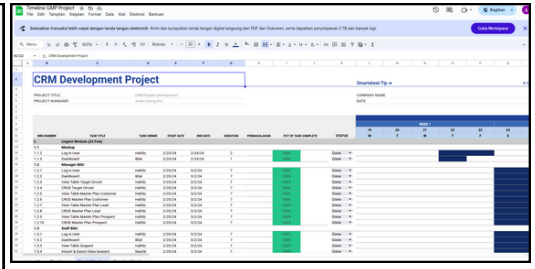
Publication Link
https://dev-crm.persadalab.com/

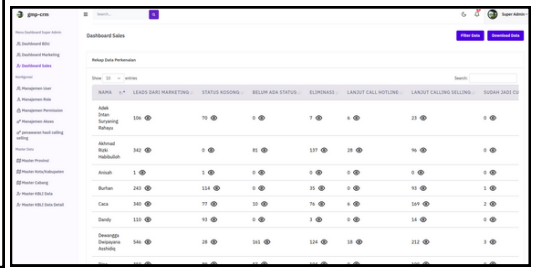
Artwork/Project Description

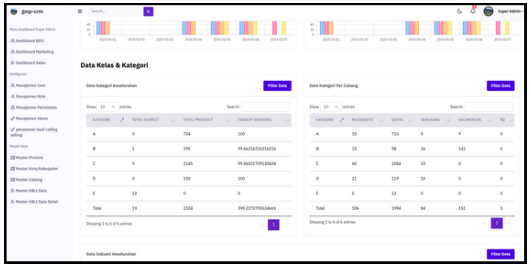
Group Project: A Custom CRM Platform for PT. Graha Mutu Persada

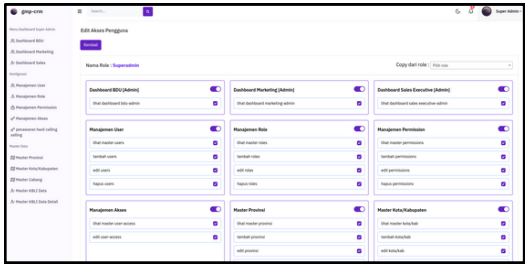
GMP CRM is a web-based Customer Relationship Management system developed for PT. Graha Mutu Persada, an environmental laboratory company based in Mojokerto, East Java. This project was a major milestone in my journey. It was my first time leading a team as a Project Manager, building a system for a large company, and handling the legal and business aspects of a freelance engagement. One of my earliest challenges was outside the technical scope: I had never created a Memorandum of Understanding (MoU), set project pricing, or negotiated with corporate stakeholders. I quickly had to learn how to draft professional agreements, define scopes clearly, estimate fair pricing, and communicate with confidence. This experience taught me essential lessons about professionalism, negotiation, and managing client relationships. On the technical side, the project involved handling over 700,000 rows of customer and transaction data from Excel files. I underestimated the effort of data cleaning, which became a major hurdle due to inconsistencies, duplicates, and missing fields. I built a custom data processing pipeline to handle it effectively. To maintain performance with such a large dataset, I implemented optimized queries, pagination, and asynchronous data loading. These efforts ensured the system remained fast, scalable, and user-friendly, capable of supporting both internal operations and future data growth. Managing the development team was another huge learning curve. With six developers under my leadership, I was responsible for dividing tasks, setting clear deadlines, coordinating workflows, and ensuring everyone was aligned with the project's goals. This required me to develop clear communication habits, project tracking systems, and conflict resolution strategies, all while maintaining progress toward our delivery milestones. I also had to communicate regularly with high-level stakeholders, including the CEO, CTO, CMO, and directors, which taught me how to present updates, justify decisions, and respond diplomatically to feedback and change requests. Another important lesson came from a mistake I made during development. At one point, I neglected to back up the database and failed to implement a soft-delete mechanism. This resulted in the accidental loss of important company data, a serious error that led to me being directly reprimanded. It was a humbling moment that taught me the importance of redundancy, safety nets, and the trust clients place in developers to protect their systems. In conclusion, leading the GMP CRM project was a deeply transformative experience. I grew not just as a developer, but also as a team leader, negotiator, communicator, and problem-solver. The experience pushed me far beyond technical work and into the world of real-world responsibility, where delivering a solution also means navigating people, expectations, and high-stakes decisions.

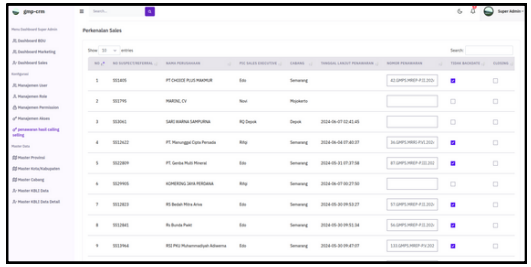












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Portfolio Submission for
Apple Developer Academy Cohort 2026

Artwork/Project Title
Digital Odyssey

Year Accomplished
2024

Role/Position
Fullstack Developer

Publication Link
https://digital-odyssey-game.vercel.app/

Artwork/Project Description

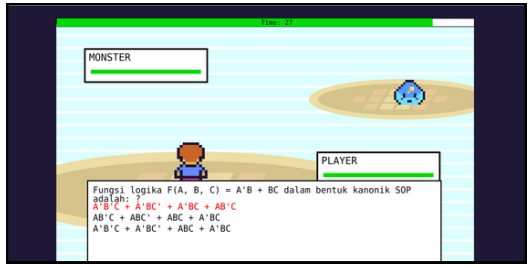
Individual Project: A 2D RPG Game Website to Boost Student Learning Motivation

Digital Odyssey is a web-based 2D RPG game I developed as part of a class assignment, where the challenge was to create an application that could enhance student motivation through gamification. Inspired by this theme, I envisioned a game that would not only entertain but also ignite a sense of progress, exploration, and achievement, mirroring the journey of learning itself. What began as a simple concept quickly evolved into one of the most ambitious and rewarding projects I've ever built.

With no prior experience in game development, I dove deep into research to understand the landscape. I explored narrative structure, motivational design, game architecture, and technical frameworks. I evaluated several tools before deciding to use Kaboom.js (now Kaplay.js), a lightweight JavaScript game engine tailored for building retro-style 2D RPGs. To manage map design, I used Tiled Map Editor to create interactive environments and level layouts. Meanwhile, I handled authentication, user data, and dashboard UI using Next.js and NextAuth, and connected everything through custom APIs and secure database access.

One of my biggest technical challenges was integrating Kaplay.js with Next.js. Since the two frameworks are not natively compatible, I engineered a creative solution: separating the game and the web app into two standalone projects and embedding the game within the main application using an iframe. API communication between the two layers was handled securely via HTTP requests, allowing real-time progress tracking, scoring, and gameplay events to be stored in the backend.

Beyond the technical hurdles, I spent weeks developing the game's storyline—crafting an engaging journey that rewards players for completing challenges modeled after real-life study milestones. I sourced and adapted open-source assets, including background music, tilesets, and sprite sheets to give the game a rich, nostalgic atmosphere. I also created supporting documentation, including ERD, activity diagrams, and use case diagrams, to clearly communicate the system's logic and scalability. The biggest challenge, however, was managing time. With only two months and no background in game dev, I had to absorb vast amounts of new knowledge while building a fully functional prototype. I watched hours of tutorials, read Medium articles, joined developer forums, and pushed through countless iterations—all while balancing my academic responsibilities. Despite the hurdles, Digital Odyssey came to life and exceeded expectations. It earned praise from my lecturer and was recognized as one of the best projects in the class. My lecturer even recommended submitting it to a national-level competition, a moment that validated all the late nights, problem-solving, and personal growth I experienced during the process. In conclusion, Digital Odyssey taught me how to learn fast, adapt to new domains, and push creative and technical boundaries under pressure. It wasn't just a class assignment—it was a journey into unknown territory that sharpened my curiosity, resilience, and ability to bring an idea from vision to reality.



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Portfolio Submission for
Apple Developer Academy Cohort 2026

Artwork/Project Title
Diflex

Year Accomplished
2025

Role/Position
Team Leader & Fullstack Developer

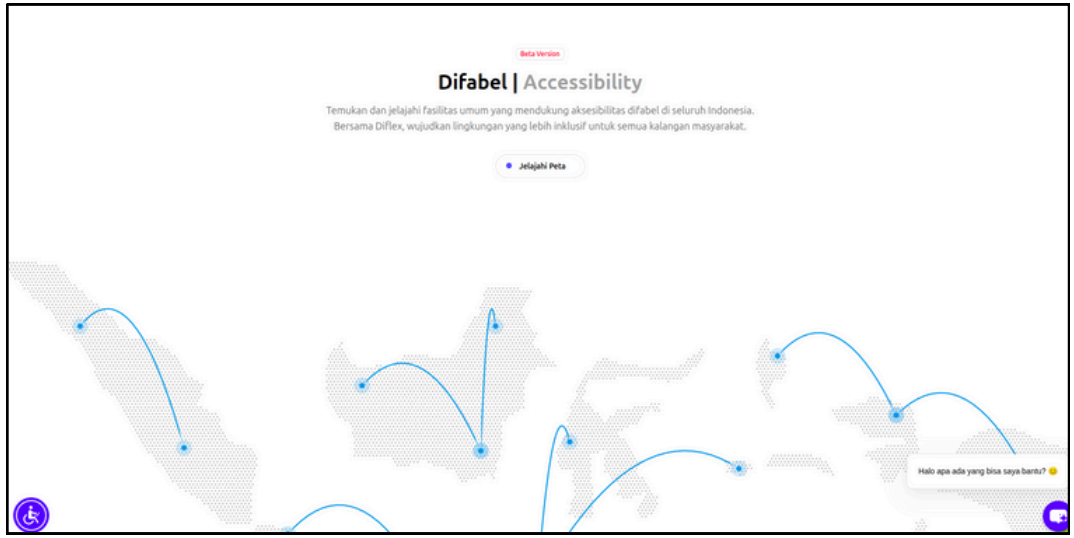
Publication Link
https://diflex.vercel.app/

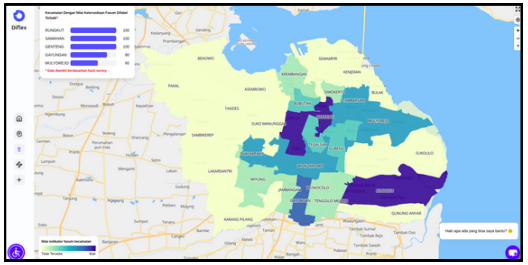
Artwork/Project Description

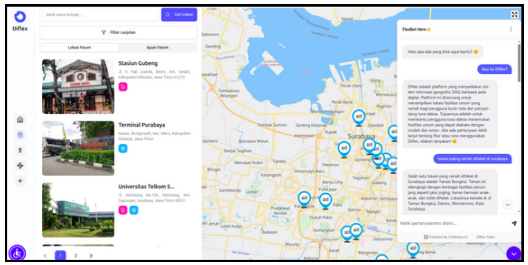
Group Project: A GIS-Based Accessibility Mapping Platform for People with Disabilities

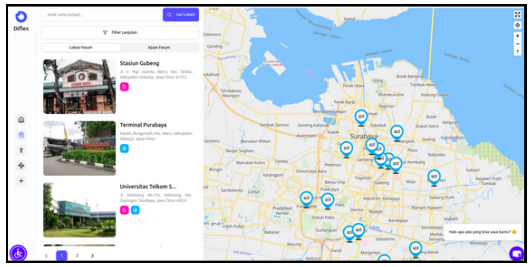
Diflex, short for Difabel Explorer Flexibility, is a GIS-powered web application we built for the MAPID Competition 2025. The idea behind this project was born from a personal struggle: one day, I took my grandfather, who lives with a physical disability (tuna daksa), out for a walk, only to realize how inaccessible many public places are. Ramps were missing, accessible restrooms didn't exist, and sidewalks were far from friendly. That moment sparked a bigger question in my mind: How many other people face the same invisible barriers every day, and how can technology help? From that point of empathy, Diflex was born. Our mission was to create a digital map of Surabaya that visualizes which areas already have disability-friendly public facilities and allows users to contribute by suggesting improvements or new installations. By crowdsourcing accessibility data, we aimed to bridge the gap between urban development and inclusivity, while also providing a valuable tool for decision-makers and everyday users alike. As Team Leader and Full-Stack Developer, I was responsible for making key technical decisions, designing the database structure, building the entire frontend, and contributing to backend development. We chose Next.js for frontend development, Prisma as our ORM, and NeonDB as the database layer. One of the biggest challenges we faced was the fact that no one in our team had any prior experience with GIS (Geographic Information Systems). I had never heard of PostGIS before this project, yet I had to quickly learn how spatial data works, how to structure it efficiently, and how to integrate it with a modern web stack. I spent days studying spatial queries, experimenting with tile layers, and figuring out how to combine geospatial functionality with interactive UI components. After trial and error, I managed to integrate GIS capabilities into our app and make it intuitive for users.

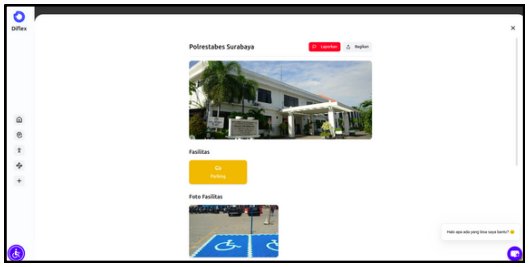
Another challenge was ensuring the site was accessible for visually impaired users, particularly those with low vision or blindness. I integrated UserWay, a third-party accessibility tool, to enhance the experience for screen reader users and improve contrast, font scaling, and keyboard navigation, features often overlooked but critical for inclusivity. To add greater value and encourage data-driven action, I introduced a unique feature: the Accessibility Index. This index analyzes the availability of accessible infrastructure in each district of Surabaya and visualizes it through thematic mapping, enabling local governments and NGOs to make more informed decisions. Even though we didn't win the competition but our team finished in 6th place due to a weak business model, this project taught me more than I expected. I learned how to lead a cross-functional team, how to learn and implement a completely new technology domain (GIS) in a limited time, how to think about inclusivity from both a design and technical perspective, how to communicate with users of all kinds, and how to transform personal frustration into meaningful solutions. In the end, Diflex was more than a competition entry. It was a project that helped me grow as a leader, engineer, and advocate for inclusive design. It reminded me why I love building technology, because even a simple map can open doors literally and metaphorically for someone who has long been locked out.

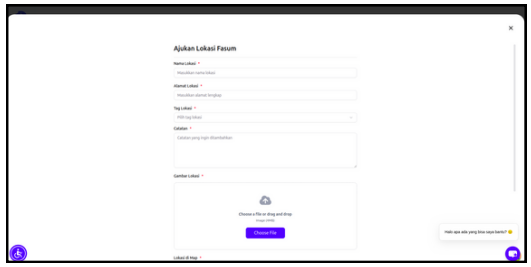












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Portfolio Submission for
Apple Developer Academy Cohort 2026

Artwork/Project Title
Motoflex

Year Accomplished
2025

Role/Position
Project Manager & Fullstack Developer

Publication Link
https://github.com/Aswin-art/motoflex

Artwork/Project Description

Group Project: A Mobile App for Online Spare Part Booking and On-Demand Technician Services

Motoflex was a project conceived during my internship, where we were challenged to create a mobile application with full creative freedom. Together with my team, which consisted of a UI/UX engineer, a backend developer, and myself, I proposed building a platform where users could seamlessly order motorcycle spare parts or request technician services online, all through a single mobile app. The idea was well-received, and even the CEO of the company where I interned expressed strong support, promising to help find potential investors to bring our concept to life.

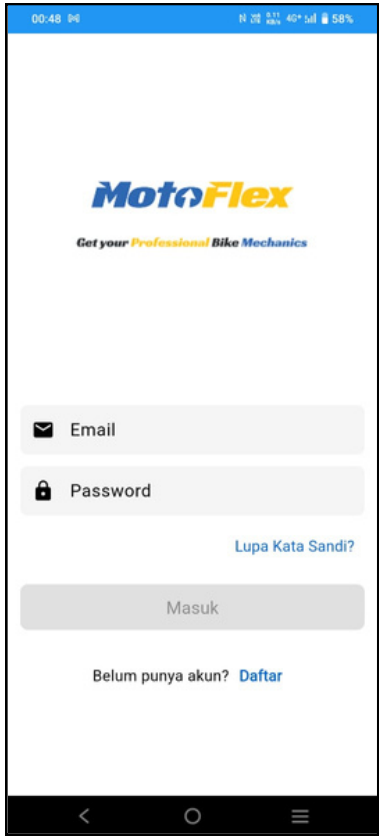
In this project, I took on the dual role of Project Manager and Full-Stack Developer. As a project manager, I led the entire development cycle: breaking down features, assigning tasks, setting deadlines, and ensuring timely delivery using Jira for task management and team coordination. This was my first time managing a mobile app development process, and it pushed me to grow rapidly as both a technical and strategic leader.

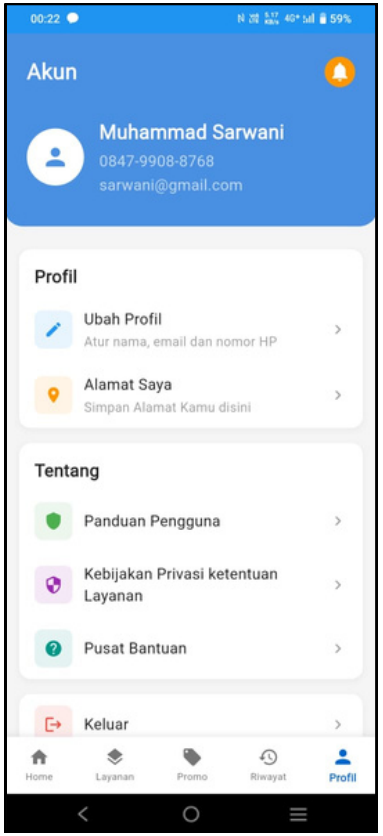
Despite my background in web development, I had never built a mobile app before. This presented a steep learning curve. I spent weeks researching and experimenting before ultimately deciding on Flutter as our framework due to its cross-platform capabilities and strong developer ecosystem. Beyond the UI, I also contributed significantly to the backend, helping develop and structure RESTful APIs using Laravel, and ensuring they were optimized for mobile consumption. This included managing authentication flows, securing data transactions, and testing integrations between Flutter and the backend.

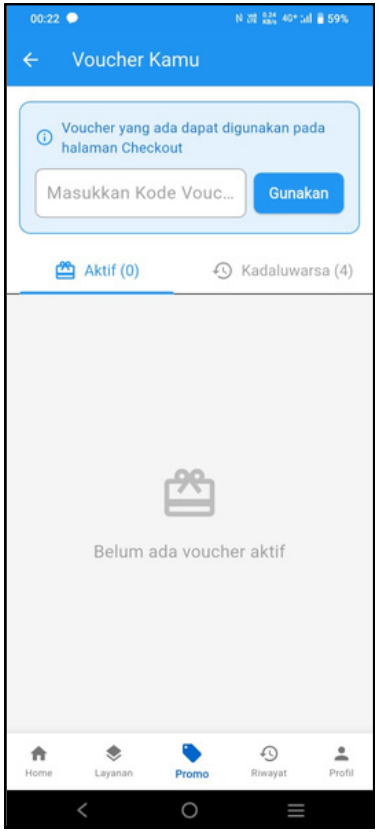
Another challenge was choosing an architecture that would support clean code and scalability. I decided to implement Clean Architecture, which taught me how to better separate concerns between layers (UI, domain, and data), and gave the team a solid foundation for future development. I also took part in code reviews, backend logic implementation, and debugging issues that arose during testing across devices.

What made this project particularly rewarding was the growth it demanded. I had to learn new tools, adapt to a new platform, collaborate across roles, and still deliver a polished, functional prototype within our internship timeframe. The moment our CEO praised our work and expressed investment interest was not just validation, it was a milestone that reflected how far we'd come.

In conclusion, Motoflex was more than a technical exercise, it was a journey into the unknown. It taught me how to lead, how to adapt quickly, and how to bring a product to life from idea to execution. It also proved to me that with the right mindset, even unfamiliar tools and platforms can become powerful instruments of creation.







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